EXERCISE - STATISTICS FOR AI
Summer Semester 2026 (Mag. Thomas Forstner)

366.591 366.592 366.593 366.594 366.595

74. The Hotel Statistic Inn has a fire alarm system which gives an alarm with a probability of 99% in the event of a fire outbreak. Occasionally, the system gives false alarms. According to the night porter, this happens four times a year (corresponds to a probability of 4/365). The probability of a fire breaking out on a certain night is 0.02%.

    Someone spends a night at the Statistic Inn and hears the fire alarm. What is the probability of a fire now? 0.018

75. A spam filter works by having users train the filter. It looks for patterns in mails marked as spam by the user. For example, it may have learned that the word “free” appears in 30% of the mails marked as spam. Assuming 5% of non-spam mails include the word “free” and 40% of all mails received by the user are spam.

    Calculate the probability that a mail is spam, if it contains the word “free”. 0.8

76. At the end of the semester, 40% of the students in a study program fail mathematics, 30% fail chemistry, and 20% fail both mathematics and chemistry. Now, a student is chosen at random.

    What is the probability that this student failed chemistry, given that he already failed mathematics? 0.5

77. Joe is randomly selected for a drug test. On average, 3% of all tested people are drug users. The drug test used correctly identifies drug users 96% of the time and correctly identifies non-drug users 93% of the time.

    Assuming that Joe has tested positive for drugs, calculate the probability that Joe really has consumed drugs. 0.298

78. A pregnancy test should be positive with a probability of 95%, if the woman is pregnant (“test indicates pregnancy”). If the woman is not pregnant, the test should be negative with a probability of 96% (“test does not indicate pregnancy”). 25% of the women who take this pregnancy test are actually pregnant.

    What is the probability that the woman is pregnant, if the test is positive? 0.888

79. A TV network wants to produce a new TV show, “CSI Statistics”. A pilot episode was produced and shown to some people. Of the viewers of this pilot episode, 65% were over 20 years. 60% of the viewers over 20 years and 50% of the rest liked the pilot episode.

    A viewer of the pilot episode “CSI Statistics”, who liked the show, is randomly selected. What are the chances that he is under 20 years? 0.31

80. A new screening test for the early detection of a specific metabolic disease in infants has been developed. This screening test does not detect the present metabolic disease in 0.03% of the cases. In 0.1% of the cases, it mistakenly detects the metabolic disease. Furthermore, it's known that on average, 100 infants out of 1.1 million infants have this metabolic disease.

    What is the probability that an infant, who was diagnosed as sick, has this metabolic disease? 0.083

81. Three different routes with different degrees of difficulty can be used to climb a mountain. Of the people who attempt to climb the mountain, experience shows that 75% use route I, 15% use route II and 10% use route III. The probability of success is 90% on route I, 70% on route II and 40% on route III.

    What is the probability that a successful climber has used route I? 0.823

82. At a gym, 80% of members are enrolled in a fitness program, and 50% of members are enrolled in a yoga program. All members are enrolled in at least one of these programs. What is the probability that a randomly selected member of this gym ...

    a) is enrolled in both the fitness program and yoga program? 0.3
    b) is enrolled in the yoga program, given that the member is enrolled in the fitness program? 0.375
    c) is enrolled in the fitness program, given that the member is enrolled in the yoga program? 0.6
    d) is enrolled in the yoga program but not in the fitness program? 0.2
    e) is enrolled in at most one of the two programs? 0.7

83. Someone wants to send a Christmas card by mail to 10 friends. 2 of these friends live in Vienna. Unfortunately, the person has only 4 stamps, so he decides to randomly select 4 out of the 10 friends and only these 4 selected friends will get a Christmas card.

    What is the probability that just one friend living in Vienna is among the 4 selected friends? 0.533

84. A continuous random variable X has the density:

$$f(x) = \begin{cases} (1 - \frac{x}{10}) \cdot k & 0 \le x \le 10 \\ 0 & else \end{cases}$$

    a) Calculate the constant k so that the probability density function is valid. k = 0.2
    b) State the cumulative distribution function. F(x) = x/5 - x²/100
    c) Calculate: P(X > 9) 0.01
    d) Calculate: P(|X – 4| ≤ 3) 0.72
    e) Calculate: P(X = 3) 0

85. Suppose that the error in the reaction temperature, in degree Celsius, for a controlled laboratory experiment is a continuous random variable X having the probability density function below:

$$f(x) = \begin{cases} \frac{x^2}{3} & -1 < x < 2 \\ 0 & else \end{cases}$$

a) Calculate: P(X < 0.7) 0.149

b) Calculate: P(–1 < X < 2) 1

86. A box contains 3 red balls and 5 blue balls. Balls are randomly drawn without replacement until a red ball is drawn. The random variable X describes the number of draws needed for a red ball (including the draw of the red ball).

State - in tabular form - the probability distribution function in the box below.

X	1	2	3	4	5	6
P(X)	0.375	0.268	0.179	0.107	0.054	0.018

87. The random variable X is defined as the sum of the numbers on 3 cards, which are drawn simultaneously from a deck of 5 cards numbered from 1 to 5.

State - in tabular form - the probability distribution function in the box below.

X	6	7	8	9	10	11	12
P(X)	0.1	0.1	0.2	0.2	0.2	0.1	0.1

Please keep the formal guidelines for submitting the homework assignments in mind to avoid losing points unnecessarily.